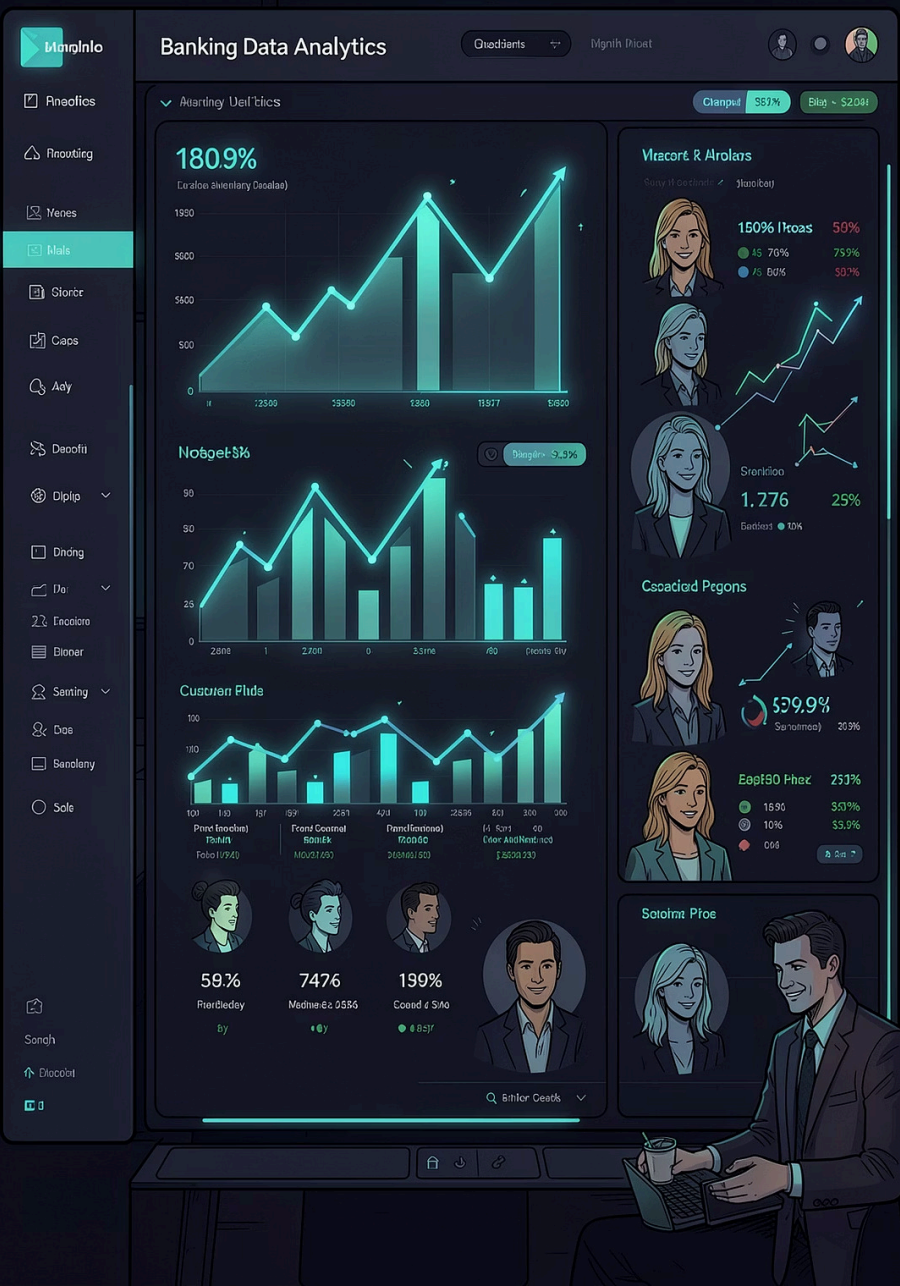




CREDIT CARD CUSTOMER CHURN ANALYSIS

Using Machine Learning to predict and prevent customer attrition in the banking sector.

MACHINE LEARNING

RANDOM FOREST

86% ACCURACY

THE CHURN CHALLENGE

THE PROBLEM

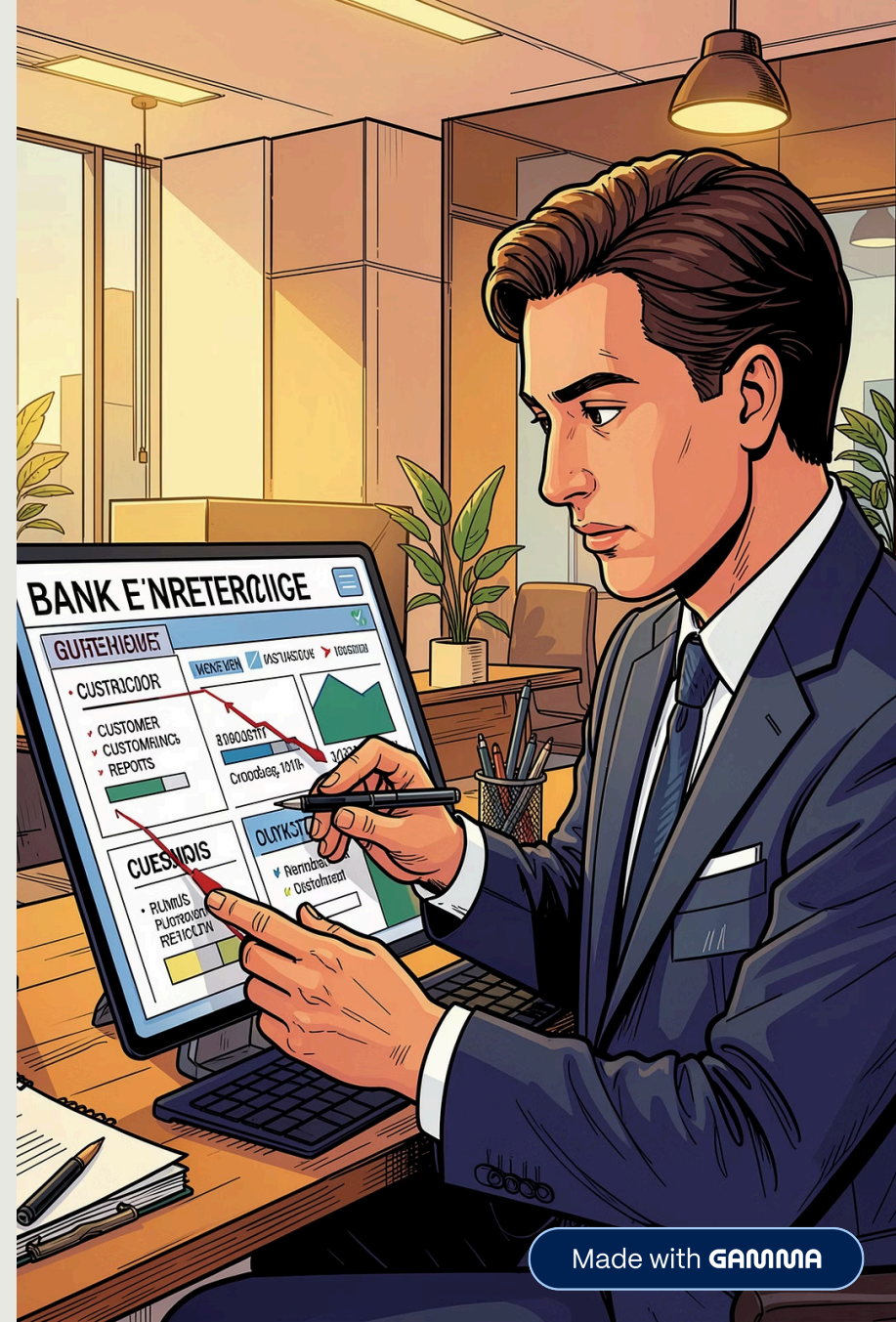
Customer churn – the loss of customers discontinuing services – significantly impacts bank profitability and long-term growth.

THE OPPORTUNITY

Banks collect vast customer data. Machine learning can identify patterns associated with attrition before customers leave.

THE GOAL

Predict whether a customer will leave based on demographic, financial, and account-related attributes.



DATASET OVERVIEW

The **Churn_Modelling.csv** dataset contains bank customer records with demographic and financial attributes.

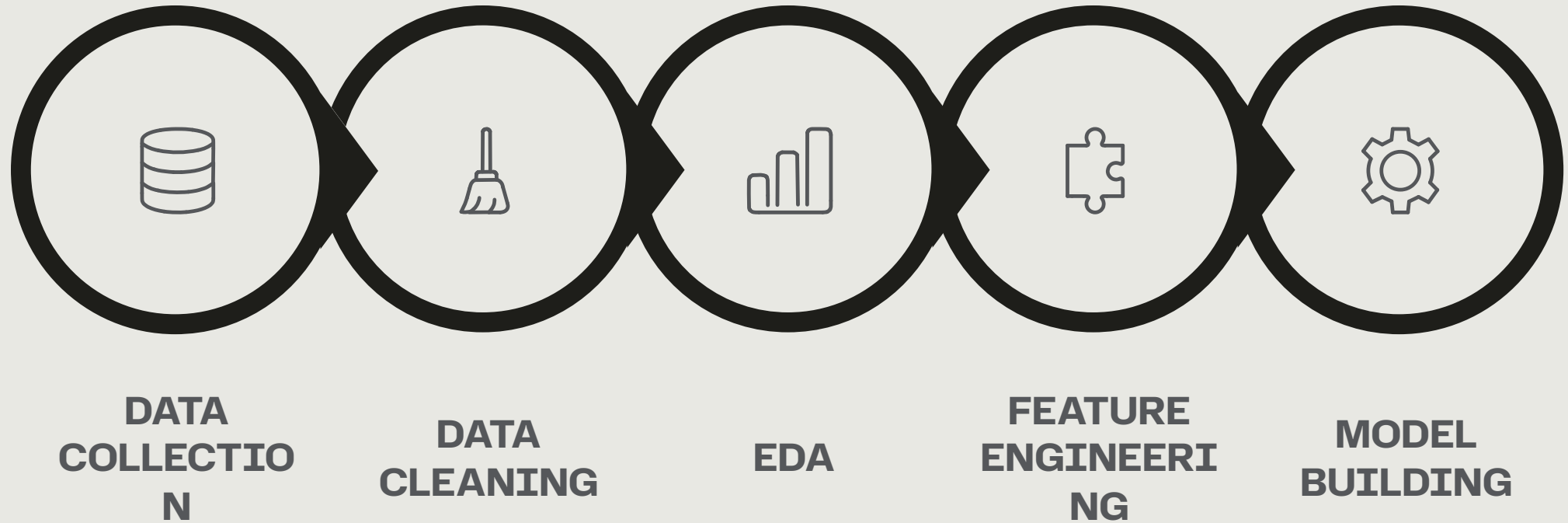
The illustration shows a desk with a green plant on the left, a notebook with a green cover on the right, and a tablet in the center. The tablet screen displays a 'CUSTOMER' data table with columns for 'CUSTOMER', 'Coads', 'Financial', 'Vraust', 'Frarda', and 'Hloc'. The table contains 10 rows of data, including customer names, codes, and various numerical values. The background is a light blue wall with a window showing a green plant outside.

CUSTOMER	Coads	Financial	Vraust	Frarda	Hloc
☐ CUS:DOMFOUSTOESN	\$,3,000				
☐ CUSTO9MER+CD2340/2R	\$300	\$269000		\$1,080	\$5,000
☐ 00S285 PMOKES	\$000	456000		\$1,950	\$800
☐ CUS080811+60201/E/S	\$380	\$0,5000		\$7,130	\$,90
☐ CX0810-PMUKES	\$880	25,300		\$4,080	
☐ CX023D-MBCIENN	\$200		\$4,000	-\$,38080	
☐ C1000981Y 500C1RES	4360		\$6,080	-\$12000	\$7,800
☐ CRI01M46 EDOIBN	?	\$50	\$6,900	\$15,900	\$6656
☐ CA00CA01T+R01C/C02873	\$70	4100	\$84200	-\$,9,500	\$3,980
☐ 6/0830-VCKEYS				-\$,5,800	\$1,500
☐ CUS3D-EIOERO	\$850	\$ 80	4,68800	\$05,000	683
☐ SIG300-VIOERS	1057	1190			\$,000
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Feature	Description
CreditScore	Customer credit score
Age, Gender	Demographics
Balance, Tenure	Account info
NumOfProducts	Bank products held
IsActiveMember	Activity status
EstimatedSalary	Annual salary
Exited	Target: 0=Retained, 1=Churned

METHODOLOGY

The project followed a standard machine learning workflow from raw data to predictive model.



Each stage ensured data quality and model reliability, culminating in a robust churn prediction system.

DATA PREPROCESSING

CLEANING & PREPARATION

COLUMN REMOVAL

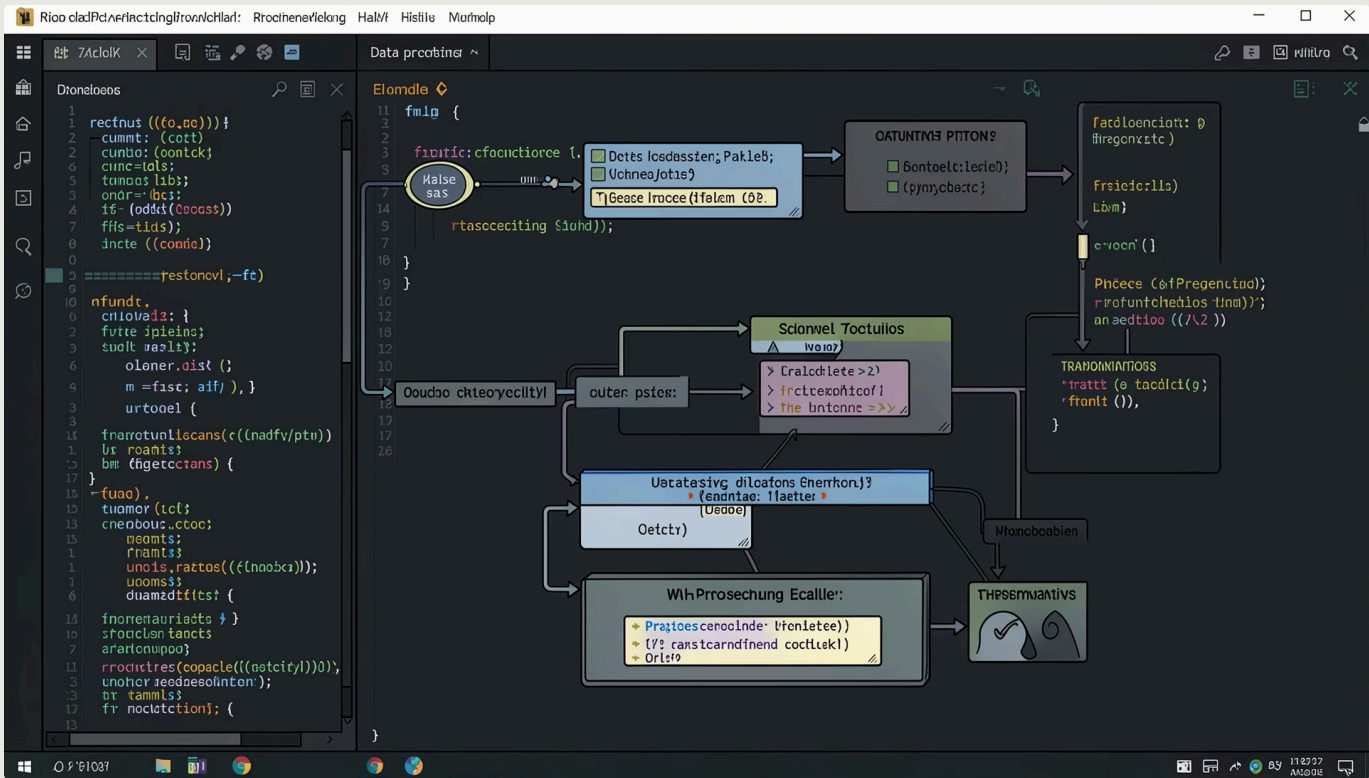
RowNumber, CustomerId, and Surname removed – irrelevant to prediction.

MISSING VALUES

No significant missing values found in the dataset.

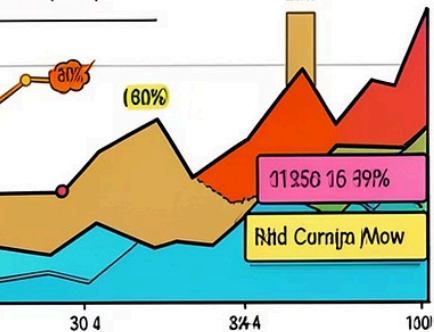
ENCODING & SCALING

Geography and Gender one-hot encoded.
Numerical variables standardized via StandardScaler.



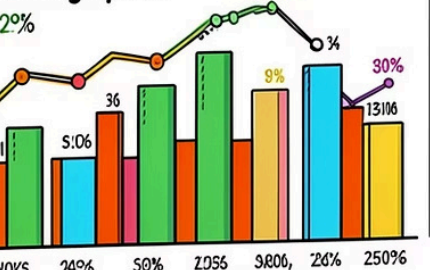
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16,50%)



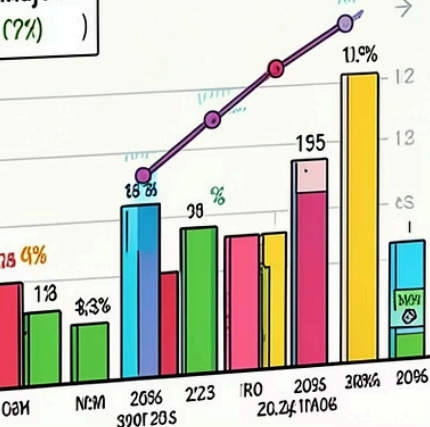
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Demographics

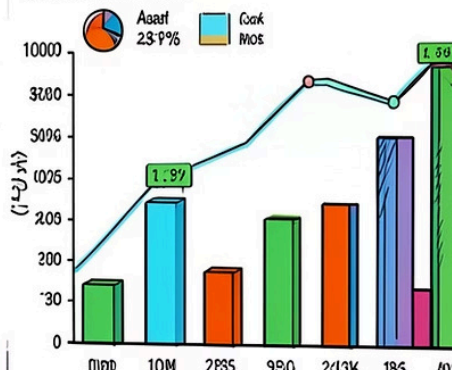


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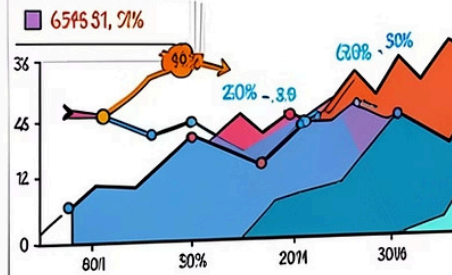


Churn Patterns

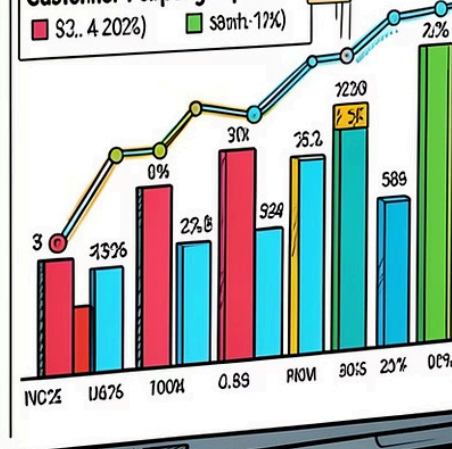


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Churn Patterns



Customer Perpongaphics



EXPLORATORY DATA ANALYSIS



CHURN DISTRIBUTION

~20% of customers churned; 80% retained. Class imbalance is typical in churn problems.



AGE FACTOR

Older customers showed a higher tendency to leave – age is a significant predictor.



GEOGRAPHY

German customers had a higher churn rate than those in France or Spain.



ACTIVITY & PRODUCTS

Inactive members and customers with fewer products were significantly more likely to churn.

MACHINE LEARNING MODELS

LOGISTIC REGRESSION

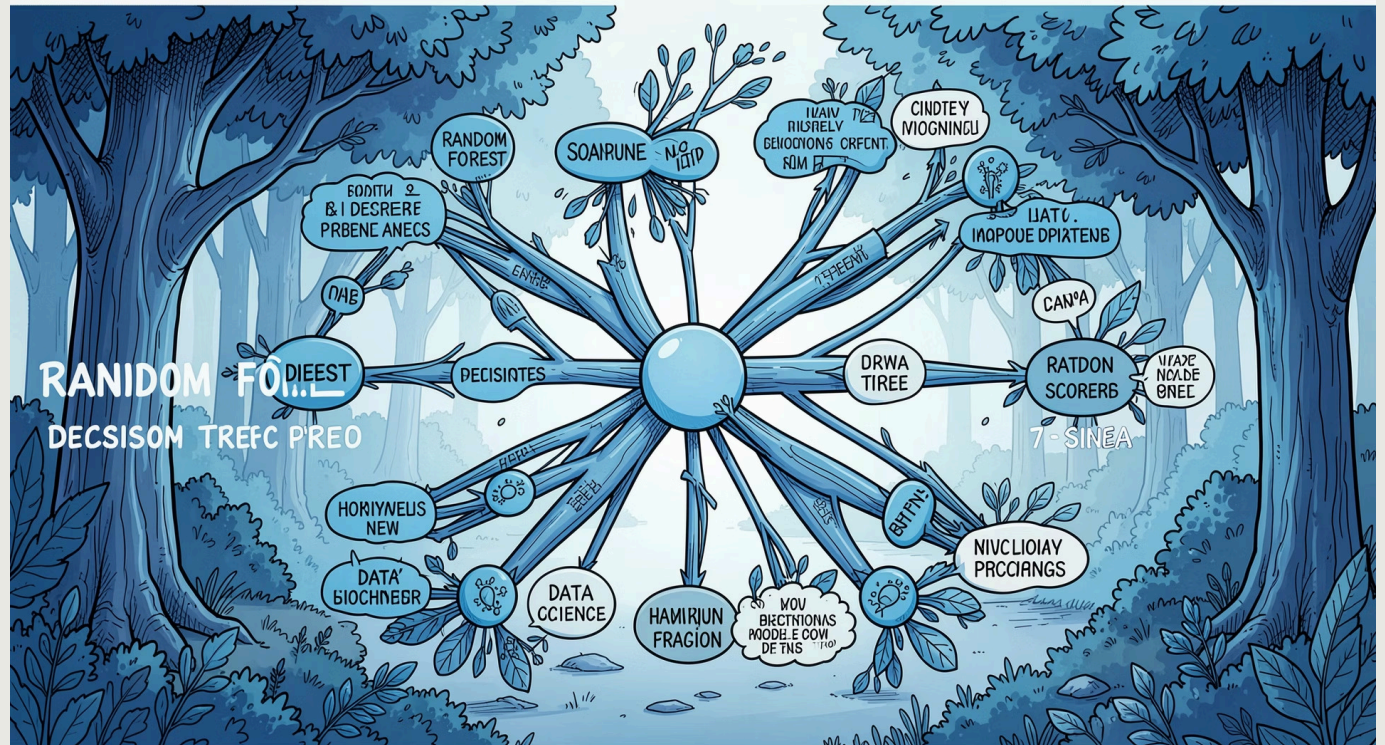
Baseline model – simple, interpretable, fast. Provides probability estimates.

~81% Accuracy

RANDOM FOREST

Primary model – handles nonlinear relationships, reduces overfitting via ensemble learning.

~86% Accuracy



✅ **Best Model:** Random Forest Classifier outperformed Logistic Regression across all evaluation metrics.

MODEL EVALUATION & FEATURE IMPORTANCE

EVALUATION METRICS

Model	Accuracy
Logistic Regression	~81%
Random Forest	~86%

Models were also assessed on Precision, Recall, F1-Score, and Confusion Matrix.

TOP CHURN PREDICTORS (RANDOM FOREST)

θ_1

AGE

θ_2

BALANCE

θ_3

NUMBER OF PRODUCTS

θ_4

CREDIT SCORE

θ_5

ISACTIVEMEMBER

θ_6

GEOGRAPHY

 **Key Finding:** Customer age and engagement level were the strongest predictors of churn.

BUSINESS INSIGHTS

⚠ HIGH-RISK PROFILE

- Older age
- High account balances
- Low product usage
- Inactive membership
- Residence in Germany

✅ LOW-RISK PROFILE

- Active account usage
- Multiple banking products
- Long-term bank relationships

📊 STRATEGIC TAKEAWAY

Cross-selling additional products and boosting engagement are key levers for improving retention.



RECOMMENDATIONS & CONCLUSION

STRATEGIC ACTIONS

→ IMPROVE ENGAGEMENT

Personalized emails, mobile notifications, and loyalty programs.

→ CROSS-SELLING

Promote savings accounts, loans, and insurance products.

→ RETENTION PROGRAMS

Target high-risk customers with special offers, fee waivers, and rewards.

→ PREDICTIVE MONITORING

Deploy the ML model for continuous churn risk monitoring and proactive intervention.

CONCLUSION

The Random Forest model achieved **~86% accuracy**, identifying age, balance, activity level, and product usage as major churn drivers. Future improvements may include **XGBoost**, **SMOTE** for class imbalance, and real-time deployment.